

- 1 (a) Define force.

.....
..... [1]

- (b) A rocket is launched vertically upwards from a launch pad. The initial mass of the rocket, with all of its fuel, is $2.83 \times 10^6 \text{ kg}$. For the first 60 s after lift-off, the rocket engine burns fuel at a rate of $2.0 \times 10^4 \text{ kg s}^{-1}$ to produce a constant thrust force.

Figure 1.1 shows the variation of the speed of the rocket with time for the first 60 s after lift-off.

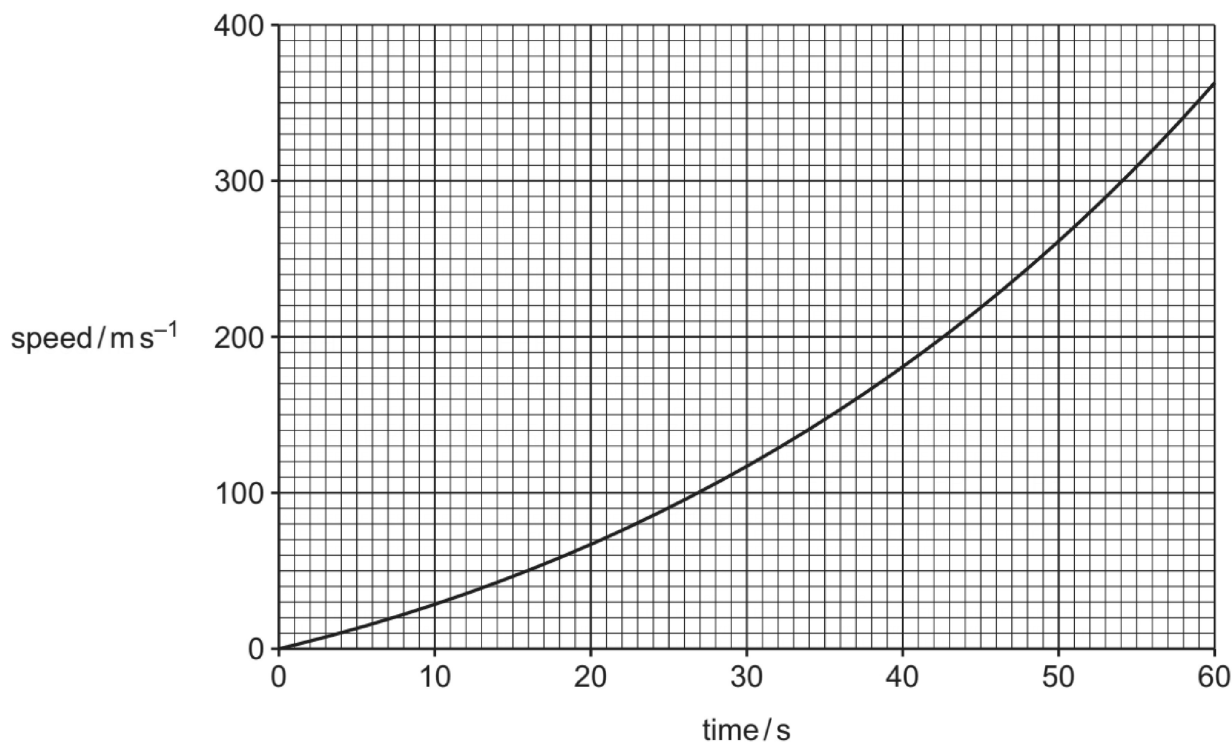


Figure 1.1

Assume that the drag force acting on the rocket is negligible over this time.

- (i) By considering the mass of unburnt fuel in the rocket, explain why the speed of the rocket does **not** increase linearly with time, despite the thrust force from the engine being constant.

.....
.....
.....
..... [2]

- (ii) Describe how Figure 1.1 may be used to determine the acceleration of the rocket at lift-off. No calculations are required.

.....
.....
..... [2]

- (iii) The acceleration of the rocket at lift-off is 2.5 m s^{-2} .

Determine a value for the thrust force T produced by the rocket engine.

$T =$ N [2]

- (iv) Determine the kinetic energy of the rocket at time 60 s after lift-off.

kinetic energy = J [3]

- (v) Use Figure 1.1 to estimate the height of the rocket above the launch pad at time 60 s after lift-off. Explain your reasoning.

height = m [2]

- (c) The mass of the emitted exhaust gases is the same as the mass of fuel burnt by the engine.

Use your answer in 1(b)(iii) to calculate the speed at which the exhaust gases are emitted from the rocket in 1(b).

speed = m s^{-1} [2]

- (d) In practice, the drag force on the rocket in 1(b) is not negligible.

By placing **one** tick (✓) in each row, complete Table 1.1 to show the effect of the drag force on the acceleration at lift-off and on the kinetic energy at time 60 s, compared with the values in 1(b) that assumed negligible drag force.

Table 1.1

	less	same	greater
acceleration at lift-off			
kinetic energy at time 60 s			

[2]

[Total: 16]